Year 10 Mathematics

Algebra 1

2.4 Solving Equations

Solving Equations

STARTER · 5 MIN

Date: _____ **Title:** Solving Equations

LEARNING OBJECTIVES

- Solve one-step and two-step equations
- Solve equations with brackets and letters on both sides
- Solve equations with fractions and from problems

STARTER *Answer in your book. No calculators.*

1. Expand $3(x - 2)$
2. Simplify $5x - 2x + 4$
3. What number $\times 4 = 28$?
4. Work out $-12 \div 3$

Starter — Answers

STARTER

Tick or correct in green pen

1. $3(x-2) =$ **$3x-6$**

2. $5x-2x+4 =$ **$3x+4$**

3. $28 \div 4 =$ **7**

4. $-12 \div 3 =$ **-4**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can solve one-step and two-step equations	☐
SILVER	I can solve equations with brackets and letters on both sides	☐
GOLD	I can solve equations with fractions and from problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Solving Equations

INTRODUCTION · 3 MIN

Do the same to both sides so the equation stays balanced.

Balance

Do the same to both sides.

$x + 4 = 9 \quad \rightarrow \quad x = 5$

Inverse

Undo + with −, and × with ÷.

$3x = 12 \quad \rightarrow \quad x = 4$

Brackets

Expand the brackets first.

$2(x + 1) = 10 \quad \rightarrow \quad 2x + 2 = 10$

Both sides

Letters on one side, numbers on the other.

$5x = 2x + 9 \quad \rightarrow \quad 3x = 9$

Examples 1 and 2 — One and Two Steps

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Solve

$3x + 5 = 20$

Example 2

Solve $\frac{x}{4} - 2 = 3$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$3x + 5 = 20$$

$$3x = 15 \quad (-5)$$

$$\mathbf{x = 5 \quad (\div 3)}$$

Tip: undo the + 5 first, then the × 3.

Example 2

$$\frac{x}{4} - 2 = 3$$

$$\frac{x}{4} = 5 \quad (+2)$$

$$\mathbf{x = 20 \quad (\times 4)}$$

Tip: undo ÷ 4 by multiplying by 4.

Task 1

PRACTICE · 9 MIN

BRONZE *Solve each equation*

1. $x + 9 = 15$

2. $5x = 35$

3. $2x - 3 = 11$

4. $\frac{x}{3} = 6$

5. $4x + 7 = 3$

★ I think of a number, multiply it by 3 and add 4. The answer is 19. Find my number.

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $x + 9 = 15, \quad x = \mathbf{6}$

2 $5x = 35, \quad x = \mathbf{7}$

3 $2x - 3 = 11, \quad 2x = 14, \quad x = \mathbf{7}$

4 $\frac{x}{3} = 6, \quad x = \mathbf{18}$

5 $4x + 7 = 3, \quad 4x = -4, \quad x = \mathbf{-1}$

★ $3n + 4 = 19, \quad 3n = 15, \quad n = \mathbf{5}$

Examples 3 and 4 — Brackets and Both Sides

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Solve

$3(2x - 1) = 21$

Example 4

Solve

$7x - 4 = 3x + 12$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$3(2x - 1) = 21$$
$$6x - 3 = 21$$
$$6x = 24$$
$$\mathbf{x = 4}$$

Tip: expand the bracket first.

Example 4

$$7x - 4 = 3x + 12$$
$$4x - 4 = 12 \quad (-3x)$$
$$4x = 16 \quad (+4)$$
$$\mathbf{x = 4}$$

Tip: take away the smaller x term from both sides.

Task 2

PRACTICE · 9 MIN

SILVER

Solve each equation

1. $2(x + 3) = 14$
2. $5(x - 2) = 30$
3. $6x + 1 = 2x + 13$
4. $4x - 5 = 7x + 7$
5. $3(x + 4) = x + 20$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $2(x+3)=14, \quad 2x+6=14, \quad 2x=8, \quad x=$ **4**

2. $5(x-2)=30, \quad 5x-10=30, \quad 5x=40, \quad x=$ **8**

3. $6x+1=2x+13, \quad 4x=12, \quad x=$ **3**

4. $4x-5=7x+7, \quad -12=3x, \quad x=$ **−4**

5. $3(x+4)=x+20, \quad 3x+12=x+20, \quad 2x=8, \quad x=$ **4**

Examples 5 and 6 — Fractions and Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5 (3 marks)

Solve $\frac{2x+3}{5} = x - 3$

Example 6 (3 marks)

The angles of a triangle are x, 2x + 10 and 3x − 40.

Find the value of x.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$\frac{2x+3}{5} = x - 3$$

$$2x + 3 = 5(x - 3)$$

$$2x + 3 = 5x - 15$$

$$18 = 3x$$

$$\mathbf{x = 6}$$

Tip: multiply both sides by 5 first.

Example 6

$$x + 2x + 10 + 3x - 40 = 180$$

$$6x - 30 = 180$$

$$6x = 210$$

$$\mathbf{x = 35}$$

Tip: angles in a triangle add up to 180°.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. Solve $5(2x - 3) = 3(x + 2)$ (3 marks)

2. Solve $\frac{x+5}{3} = 4$ (2 marks)

3. Solve $\frac{3x-1}{4} = x - 2$ (3 marks)

4. A rectangle has length $(2x + 1)$ cm and width x cm. Its perimeter is 32 cm. Find x. (3 marks)

5. Solve $\frac{x}{2} + \frac{x}{3} = 10$ (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $5(2x-3)=3(x+2),\quad 10x-15=3x+6,\quad 7x=21,\quad x=$ **3**

2. $\dfrac{x+5}{3}=4,\quad x+5=12,\quad x=$ **7**

3. $\dfrac{3x-1}{4}=x-2,\quad 3x-1=4x-8,\quad x=$ **7**

4. $2(2x+1)+2x=32,\quad 6x+2=32,\quad 6x=30,\quad x=$ **5**

5. $\dfrac{x}{2}+\dfrac{x}{3}=10,\quad 3x+2x=60,\quad 5x=60,\quad x=$ **12**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Solve $4x - 9 = 15$

SILVER Solve $8x + 3 = 5x + 18$

GOLD Solve $\frac{x+1}{2} = \frac{x+7}{4}$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $4x - 9 = 15, \quad 4x = 24, \quad x = \mathbf{6}$

SILVER $8x + 3 = 5x + 18, \quad 3x = 15, \quad x = \mathbf{5}$

GOLD $\frac{x+1}{2} = \frac{x+7}{4}, \quad 2(x+1) = x+7, \quad 2x+2 = x+7, \quad x = \mathbf{5}$

Exam Practice — Forming and Solving

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(4 marks)

(a) Solve $5x - 7 = 18$ (2)

x = ...

(b) Solve (2)

$$\frac{x}{4} + 3 = 9$$

x = ...

Question 2

(3 marks)

The perimeter of the rectangle is 36 cm.

Work out the value of x.

Diagram NOT accurately drawn



x = ...

Exam Practice — Brackets and Angles

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(3 marks)

Solve

$4(2x - 3) = 3x + 13$

x = ...

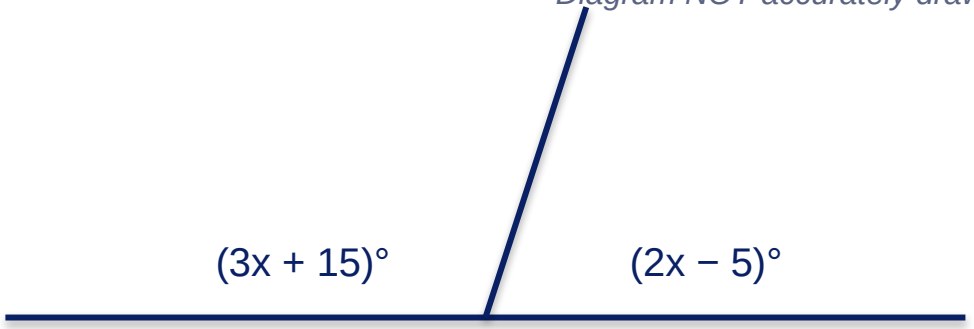
Question 4

(3 marks)

The diagram shows angles on a straight line.

Work out the value of x.

Diagram NOT accurately drawn



x = ...

Exam Practice — Fractions and Geometry

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(3 marks)

Solve

$$\frac{3x + 1}{4} = \frac{x - 3}{2} + 2$$

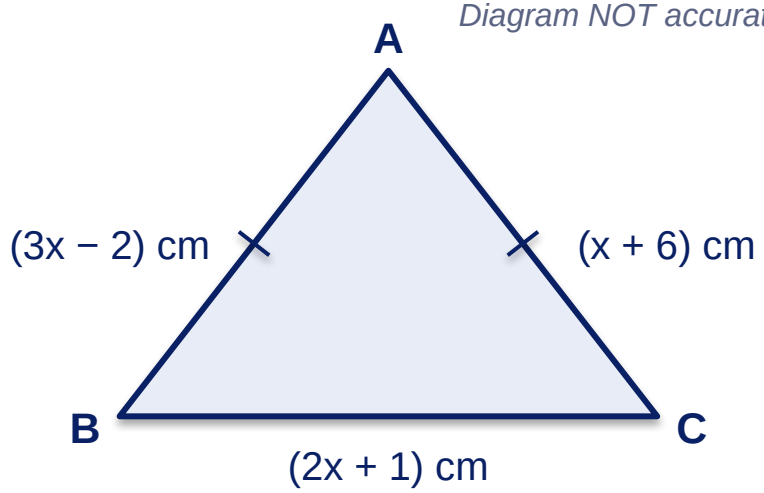
x = ...

Question 6

(4 marks)

Triangle ABC is isosceles with AB = AC.
Work out the perimeter of the triangle.

Diagram NOT accurately drawn



..... cm

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $5x = 25$ — M1

$x = 5$ — A1

(b) $\frac{x}{4} = 6$ — M1

$x = 24$ — A1

Tip: undo + and − first, then × and ÷.

Question 2

$2(x + 4) + 2(2x - 1) = 36$ — M1

$6x + 6 = 36$ — M1

$x = 5$ — A1

Tip: form the equation from the perimeter.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$8x - 12 = 3x + 13$ — M1

$5x = 25$ — M1

$x = 5$ — A1

Tip: expand first, then collect x on one side.

Question 4

$3x + 15 + 2x - 5 = 180$ — M1

$5x + 10 = 180$ — M1

$x = 34$ — A1

Tip: angles on a straight line add to 180°.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

× 4: $3x + 1 = 2(x - 3) + 8$	M1
$3x + 1 = 2x + 2$	M1
$x = 1$	A1

Tip: multiply EVERY term by 4.

Question 6

$3x - 2 = x + 6$	M1
$x = 4$	A1
$10 + 10 + 9$	M1
$= 29$ cm	A1

Tip: equal sides give the equation.